Introduction

Understanding of Cisco Command Line Interface (CLI) Command Modes is a fundamental skill for the FSNA certification . This document will cover and explain the topic

Basically, CLI Command Modes define levels of access to the router or switch operating system and what you can see and configure within it

Each command mode is identified with a symbol or letters and a symbol after the device name. Here is a brief overview of the three main Command Modes, the device name (Router) is just for teaching purposes

Command Mode Overview

User Exec Mode: `Router>`
Router> here, we see the device name “Router”, and a “greater than” symbol “>” at the end . This is the lowest level of access

Privileged Exec Mode : `Router#`
Router# the device name is the same “Router”, but now we have a pound sign “#” at the end . This is where most “show” commands are executed (commands that let you examine the configuration on the device)

Global Configuration Mode: `Router(config)#`
Router(config)# same device name , “Router”, but this time we see “(config)#” at the end. This mode allows us to make changes in the device configuration or access specialized configuration modes .

Changing Command Modes

The levels of the Command Mode, from lowest level of access to the highest:
1 User Exec Mode >

2 Privileged Exec Mode #
3 Global Config Mode (config)#
4 Configuration sub modes - there are several of these

1 To move UP from User Exec to Privileged Exec mode, type “enable”

```
Router>enable
Router#
```

2 To move UP from Privileged Exec to Global Configuration mode, type “conf t”

```
Router#conf t
Router(config)#
```

3 To move DOWN in any mode (except from Privileged Exec to User Exec , which you will hardly ever do) type “exit”

```
Router(config)#exit
Router#
```

Other Configuration Command Modes

There are several other modes accessed from Global Configuration Mode, I will include a few as examples. Each one will will have a prompt with the word “config” within it .

`Router(config-if)#` Interface Configuration Mode

`Router(config-line)#` Line Configuration Mode

`Switch(config-vlan)#` VLAN Configuration Mode

These are all accessed from Global Configuration Mode. You can exit them and return to Global Configuration Mode by typing “exit”

Working with Configuration Modes in the CLI

It is critically important that when doing labs, you note the Configuration Mode listed in the Lab guide. If you enter a command in the wrong Configuration mode, you will get an error message saying

% Invalid input detected at ‘^’ marker.

Example:

```
Switch#in fa0/10
          ^
% Invalid input detected at '^' marker.
```

Labs will have instructions that indicate the Command Mode that one must be in to enter a command properly
Examples:
(config)# hostname ICND1-RTR - here you are in Global Configuration Mode. You will enter the Global Configuration Mode in the CLI and type “hostname ICND1-RTR”. Do NOT type “(config)# hostname ICND1-RTR “, or you will get an error message

Router# show run - here you are in Privileged Exec Mode. You will enter the Privileged Exec Mode in the CLI and type “show run”. Do NOT type “Router# show run” or you will get an error message